

Smart Weather Station: Weekly Temperature Analyzer

C Programming Lab Assessment | Topic: One-Dimensional Arrays | Level: Medium

Background

A district weather station in Karnataka records the maximum temperature once a day. At the end of each week, the station officer enters the readings into a small C program. The program analyzes them and compares them with last week's data, which is already stored in the program. Your job is to build this analyzer.

Requirements

1. Static initialization. Declare an array `lastWeek[7]` and initialize it at compile time with last week's temperatures:

```
float lastWeek[7] = {31.5, 32.0, 33.2, 30.8, 29.9, 34.1, 33.0};
```

Also declare a statically initialized array of day names:

```
char days[7][4] = {"Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"};
```

2. Dynamic (runtime) initialization. Declare `thisWeek[7]` without initial values. Read its 7 readings from the user at runtime using `scanf()`. Reject any reading outside the range 0-55 °C and ask for that day's value again.

3. Functions. Each function must take an array as an argument.

Function prototype	Task
<code>void readTemps(float t[], int n)</code>	Read and validate n readings
<code>void display(float t[], int n)</code>	Print day-wise readings in a neat table
<code>float average(float t[], int n)</code>	Return the average temperature
<code>int hottestDay(float t[], int n)</code>	Return the index of the maximum temperature
<code>int countAbove(float t[], int n, float limit)</code>	Count days above a user-given limit
<code>void compareWeeks(float cur[], float prev[], int n)</code>	Print the day-wise difference and "↑ Warmer", "↓ Cooler" or "= Same"

4. Retrieving elements. After the analysis, ask the user for a day number (1-7) and print both weeks' readings for that day using the index. Handle invalid day numbers gracefully.

5. Menu-driven program. Wrap everything in a menu that repeats until the user chooses Exit.

Sample Interaction

```
Enter temperature for Mon: 32.4
Enter temperature for Tue: 60
Invalid! Enter a value between 0 and 55: 33.1
...
----- MENU -----
1. Display this week
2. Average temperature
3. Hottest day
4. Count days above a limit
5. Compare with last week
6. Look up a specific day
7. Exit
Enter choice: 3
Hottest day: Sat (35.6 °C)

Enter choice: 5
Day   Last   This   Change
Mon   31.5   32.4   +0.9   ↑ Warmer
Tue   32.0   33.1   +1.1   ↑ Warmer
...
```